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Practical 9

AIM:- To implement the K-Mean clusturing algorithm and analyze the clusturing result using appropriate evaluation Technique

```
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
```

```
from google.colab import drive
drive.mount("/content/drive")
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=Tr
```

```
path= "/content/drive/MyDrive/ML_Dataset/Mall_Customers.csv"
df=pd.read_csv(path)
df.head(5)
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
X = df.iloc[:, [3, 4]].values
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

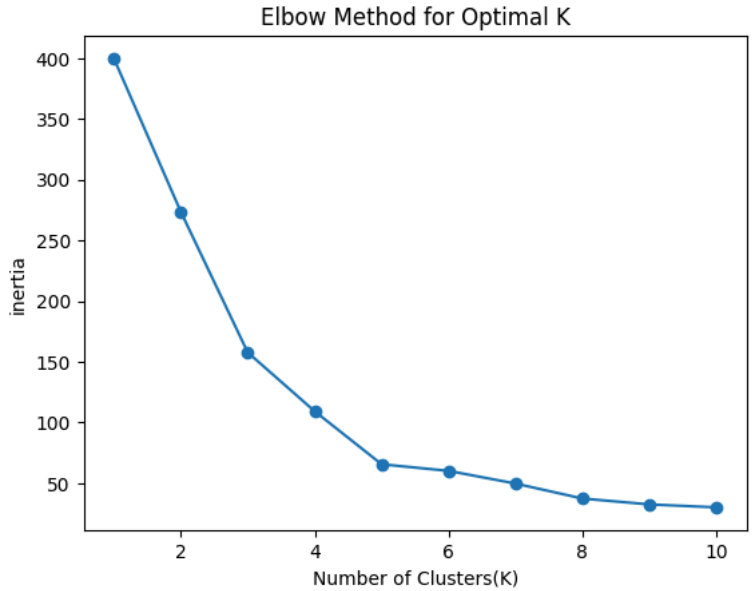
```
inertia_values = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertia_values.append(kmeans.inertia_)

print(inertia_values)

plt.figure()
plt.plot(range(1,11), inertia_values, marker='o')
plt.title('Elbow Method for Optimal K')
plt.xlabel('Number of Clusters(K)')
plt.ylabel('inertia')
plt.show()
```

```
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681, 60.132874871934206, 49.6
```




```
kmeans = KMeans(n_clusters=6, random_state=42)
clusters = kmeans.fit_predict(X_scaled)

# Add cluster labels to dataset
df['Cluster'] = clusters
(df.head())
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Cluster
0	1	Male	19	15	39	5
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	5

```
print("Inertia:",kmeans.inertia_)

sil_score = silhouette_score(X_scaled, clusters)
print("Silhouette Score:", sil_score)
```

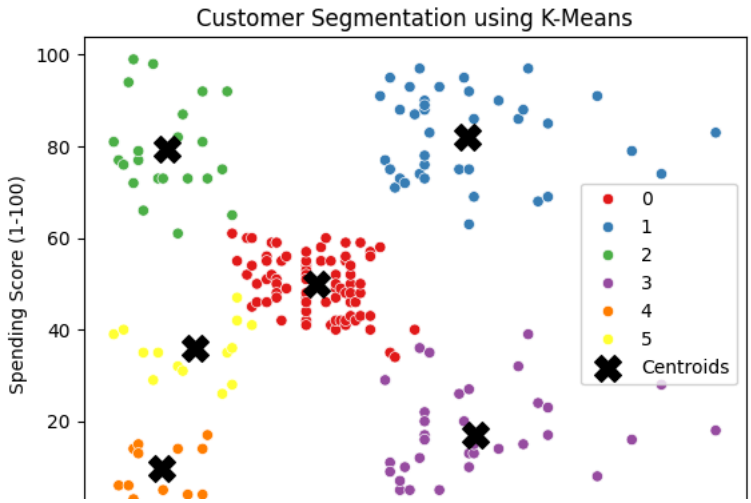
```
Inertia: 60.132874871934206
Silhouette Score: 0.5138257534676561
```

```
plt.figure()
sns.scatterplot(x='Annual Income (k$)',
                y='Spending Score (1-100)' , hue='Cluster',
                data=df, palette='Set1'
)

centroids = scaler.inverse_transform(kmeans.cluster_centers_)

plt.scatter(
    centroids[:, 0],
    centroids[:, 1],
    s=200, c='black', marker='X',
    label='Centroids'
 )

plt.title("Customer Segmentation using K-Means")
plt.legend()
plt.show()
```


Customer Segmentation using K-Means
Spending Score (1-100)
100
80
60
40
20
0
1
2
3
4
5
Centroids